Predicting Customer Order Value

for an E-Commerce Platform

Business Analytics — Project Presentation

Group 13

Nguyen Huy Hoang 20251325M
Nguyen Luong Quynh Trang 20252761M
Tran Tien Dung 20252574M
Tran Manh Tien 20252762M
Bui Duc Phan Hoang 20252004M

July 24, 2026

Brazilian E-Commerce Public Dataset (Olist)

Agenda

Problem Statement

Business Problem

The challenge: E-commerce platforms rely on *population-level averages* for operational decisions, obscuring substantial variation between individual orders.

Operational decisions affected:

- Shipping tier selection
- Upsell recommendation timing
- Fraud review prioritisation
- Promotional targeting

Objective

Replace population-level averages with **individual-level predictions** of order value.

Research Question

Given a customer's profile and session behaviour, what is the expected order value, and which factors drive it?

Business Value

Accurate order value predictions enable three categories of action:

Upsell	Shipping	Promotions
Dynamic	**Optimised**	**Targeted**
Calibrated to predicted value tier	Matched to order magnitude	Directed to highest incremental revenue

Projected Monthly Return

R$16,140/month (∼R$193,680 annually) from dynamic upsell profit and insurance compliance savings, based on 10,000 monthly orders at R$138 AOV.

Data Strategy

Data Foundation: Olist Dataset

Nine relational tables:

- 99,441 orders
- 112,650 line items
- 32,951 products
- 3,095 sellers
- 1,000,163 geolocation records

Period: September 2016 – October 2018

Critical Gap

Olist captures *what* was purchased but not *how* the customer arrived at that purchase.

No session data, clickstream, demographics, or loyalty information.

Synthetic Behavioural Data

Three tables generated following an ERD-aligned schema:

Table	Rows	Contents
`customer_profile`	96,096	Demographics, income, loyalty tier, device preference
`sessions`	99,441	Device, browser, OS, traffic source, session timing, coupons
`session_activity`	763,850	Clickstream: page views, searches, cart additions, checkout

Design Principle: Value-Conditioned Generation

Each synthetic feature is conditioned on the real order value via $v_s = z\text{-score}(\log(1 + y))$, ensuring genuine predictive signal—not random noise.

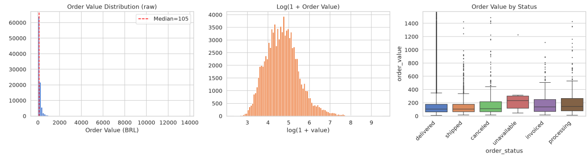
Value-Conditioned Correlation Design

Feature	Target Effect	Rationale
Monthly income	$r \approx 0.51$	Higher income $\rightarrow$ higher spending capacity
Loyalty tier	$2.5\times$ AOV spread	Loyal customers trust platform, spend more
Traffic source	Email $+64\%$ vs Google	Email $=$ loyal repeat; Google $=$ acquisition
Device type	Desktop $+7\%$	Desktop $=$ deliberate shopping, larger carts
Session engagement	$r \approx 0.2$	More deliberation $\rightarrow$ higher-value orders
Gender, education	$r \approx 0$ (indep.)	No realistic causal link to individual order value

Leakage prevention: Residual noise ($\sigma = 0.5$–0.7) ensures no single synthetic feature exceeds $r = 0.5$; model R^2 from synthetic features alone ≈ 0.10–0.25.

Exploratory Data Analysis

Target Distribution



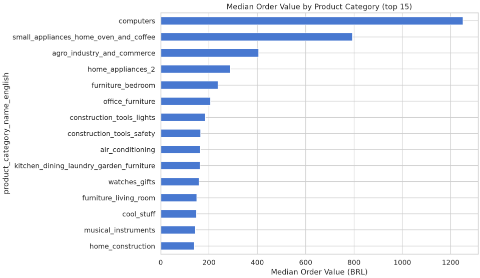
Key statistics:

- Median: R$105
- Mean: R$160
- Max: R$13,664
- Skewness: 9.37

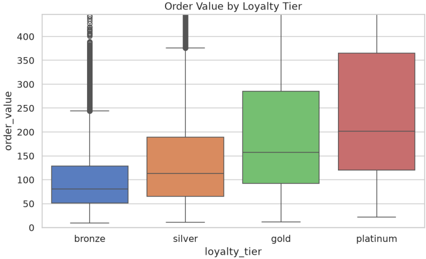
Transformation

$\log(1+y)$ reduces skewness to **0.17**, producing a near-normal distribution suitable for regression.

Value Drivers

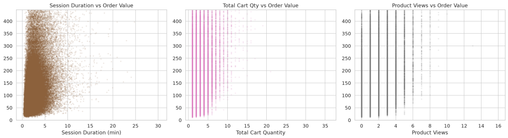


40-fold variation across categories

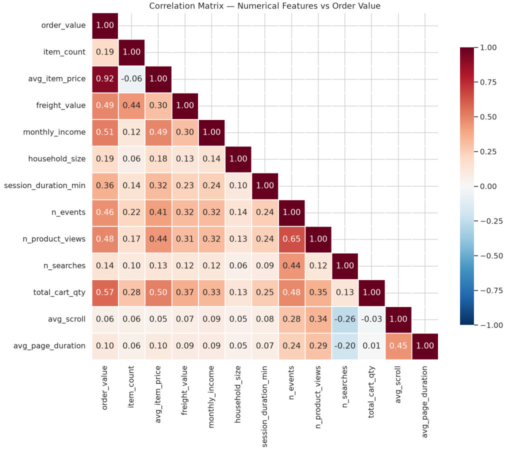


2.5$\times$ spread: bronze $\rightarrow$ platinum

Session Behavior & Multicollinearity



Session Behavior: Duration and cart additions scale with higher-value baskets.



Correlation: No independent features exceed $|r| = 0.7$, mitigating multicollinearity.

Feature Engineering

From 73 Columns to 182 Features

Group	Count	Examples
One-hot categoricals	131	Device, browser, OS, traffic, loyalty, category, seller state
Scaled numericals	47	Income, session duration, geolocation, haversine distance, RFM
Target-encoded	3	IP region, campaign name, seller state (smoothed)
Derived interactions	16	Income$\times$loyalty, cart conversion, engagement rate
Boolean flags	3	Logged-in, marketing opt-in, coupon applied
Total	**182**	VIF analysis removed 3 collinear features

Cleaning decisions:

- 1,856 cancelled/unavailable orders removed
- 71 Portuguese category names translated
- Geolocation, seller reputation, regional price indices integrated
- **No winsorization**—full value range preserved; outlier robustness via L_1 loss

Modelling

Model Comparison

Split: Time-based at 2018-04-01 (65,124 train / 32,460 test)

Model	R^2	MAE	WAPE	MedAPE	MAPE
Linear Regression	0.74	R$318	193%	27%	34%
Ridge	0.74	R$318	193%	27%	34%
Lasso	0.64	R$141	86%	32%	42%
Random Forest	0.79	R$49	30%	22%	30%
XGBoost (L_1)	**0.80**	**R$48**	**29%**	**21%**	**28%**
LightGBM (MAE)	0.80	R$48	29%	22%	29%

- L_2-based models: extreme MAE due to outlier sensitivity
- Tree-based + L_1: MAE $<$ R$50
- **XGBoost selected:** best MedAPE (21%)

Feature Importance: Validation of Synthetic Data

#	Feature	Imp.
1	income × loyalty	14.0%
2	session event count	8.8%
3	cart additions	8.8%
4	product views	5.2%
5	total cart quantity	5.1%
6	payment installments	3.5%
7	category: telephony	2.7%
8	log income	2.6%
9	category: electronics	2.4%
10	loyalty: bronze	2.3%

Key Finding

13 of 20 top features are synthetic behavioural features.

The value-conditioned generation produces data the model finds *genuinely predictive*.

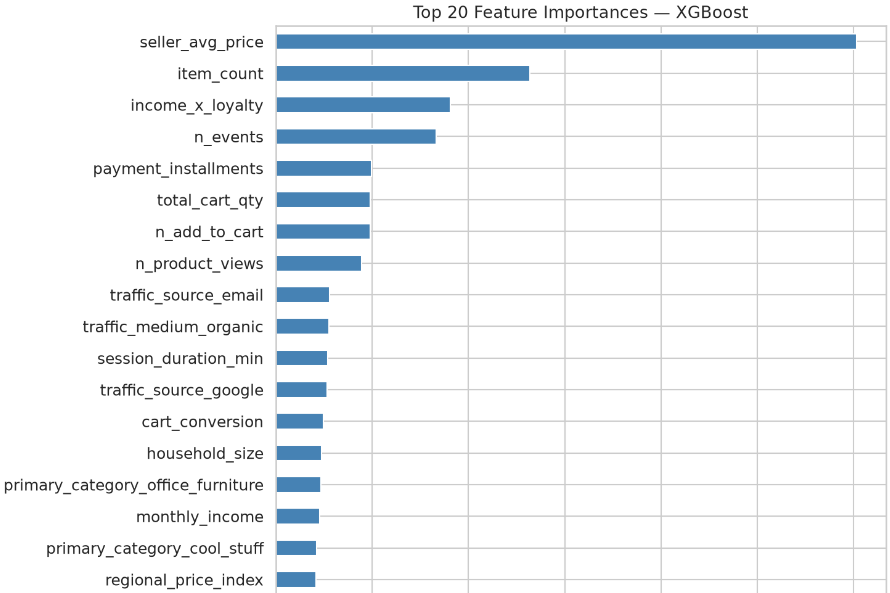
Discriminative Power

Bronze mobile, 1 item: **R$83**

Platinum desktop, 5 items: **R$520**

6.3× spread

Feature Importance (Top 20)



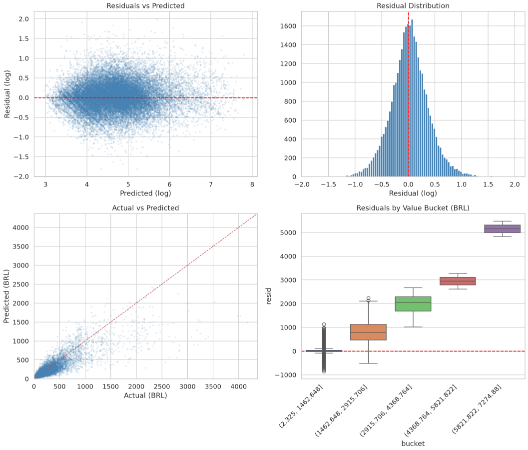
Residual Analysis

Log-Space Diagnosis

- Normal distribution of residuals
- Homoscedasticity generally maintained

Original Scale (BRL)

- Errors naturally expand for extreme high-value orders
- Validates the choice of proportional KPI (WAPE, MAPE)



Deployment & Monitoring

System Architecture

Prediction API (FastAPI)

- `POST /predict`: 30+ features → value + 80% CI
- `GET /health`: liveness check
- Model + preprocessor serialised as single artifact

Interactive Demo (Streamlit)

- Sidebar controls for all inputs
- Real-time business insights
- Standalone version for HF Spaces

Monitoring

- PSI drift detection per feature
- Evidently AI dashboards (3 reports)
- Retraining triggers:
  - Feature PSI > 0.25
  - Rolling 7-day WAPE $> 35\%$

Deployment

- Dockerfile for containerisation
- HuggingFace Spaces (Streamlit)
- `run_pipeline.sh`: single-command E2E

Business Recommendations

Actionable Recommendations

1. **Loyalty tier upgrade programmes**
   2.5× AOV spread (bronze R$82 → platinum R$202) — largest single lever
2. **Email channel investment**
   64% higher median AOV vs. Google organic; lower acquisition cost
3. **Desktop experience optimisation**
   7% higher median AOV; critical for high-value categories
4. **Real-time cart upsell triggers**
   Cart additions rank #3 in importance; target low-engagement sessions
5. **Income-tier personalisation**
   Income × loyalty = #1 feature (14%); use predicted tier for real-time segmentation

Limitations & Future Work

Limitations and Future Directions

Current Limitations

- Synthetic correlations are *designed*, not observed
- No product-level pricing features
- 97% single-order customers (limited longitudinal data)
- KPI targets not fully met

KPI	Target	Actual
MAE	$<$R$25	R$48
WAPE	$<$16%	29%
MedAPE	$<$12%	22%
MAPE	$<$15%	29%

Future Directions

- Integrate real clickstream data
- Add product-level pricing features
- Sequence models (LSTM / Transformers) for clickstream
- External data: competitor pricing, macroeconomic indicators

Path to KPI Targets

Product-level pricing + sequence architectures are expected to close the majority of the remaining gap.

Summary

Summary

Results

R^2	MAE
0.79	**R$48**
Variance explained	Mean absolute error

WAPE	MedAPE
29%	**22%**
Weighted abs. pct. error	Median abs. pct. error

Key Contributions

- Value-conditioned synthetic data with validated predictive power
- 182-feature engineering pipeline
- End-to-end reproducible system
- FastAPI + Streamlit + Docker deployment
- PSI + Evidently monitoring
- 43 tests, single-command pipeline

Bottom Line

Synthetic behavioural features drive 13/20 top predictions—validating the data generation design.

Thank You

Questions & Discussion

Repository: `github.com/alitonia/customer_value_prediction`

Live Demo: `huggingface.co/spaces/alitonia/customer_value_prediction`

Brazilian E-Commerce Public Dataset (Olist) — 99,441 orders